GIT / GITHUB

Git Cheat Sheet

GIT BASICS

`git init <directory>`	Create empty Git repo in specified directory. Run with no arguments to initialize the current directory as a git repository.
`git clone <repo>`	Clone repo located at `<repo>` onto local machine. Original repo can be located on the local filesystem or on a remote machine via HTTP or SSH.
`git config user.name <name>`	Define author name to be used for all commits in current repo. Devs commonly use `--global` flag to set config options for current user.
`git add <directory>`	Stage all changes in `<directory>` for the next commit. Replace `<directory>` with a `<file>` to change a specific file.
`git commit -m "<message>"`	Commit the staged snapshot, but instead of launching a text editor, use `<message>` as the commit message.
`git status`	List which files are staged, unstaged, and untracked.
`git log`	Display the entire commit history using the default format. For customization see additional options.
`git diff`	Show unstaged changes between your index and working directory.

UNDOING CHANGES

`git revert <commit>`	Create new commit that undoes all of the changes made in `<commit>`, then apply it to the current branch.
`git reset <file>`	Remove `<file>` from the staging area, but leave the working directory unchanged. This unstages a file without overwriting any changes.
`git clean -n`	Shows which files would be removed from working directory. Use the `-f` flag in place of the `-n` flag to execute the clean.

REWRITING GIT HISTORY

`git commit --amend`	Replace the last commit with the staged changes and last commit combined. Use with nothing staged to edit the last commit's message.
`git rebase <base>`	Rebase the current branch onto `<base>`. `<base>` can be a commit ID, branch name, a tag, or a relative reference to HEAD.
`git reflog`	Show a log of changes to the local repository's HEAD. Add `--relative-date` flag to show date info or `--all` to show all refs.

GIT BRANCHES

`git branch`	List all of the branches in your repo. Add a `<branch>` argument to create a new branch with the name `<branch>`.
`git checkout -b <branch>`	Create and check out a new branch named `<branch>`. Drop the `-b` flag to checkout an existing branch.
`git merge <branch>`	Merge `<branch>` into the current branch.

REMOTE REPOSITORIES

`git remote add <name> <url>`	Create a new connection to a remote repo. After adding a remote, you can use `<name>` as a shortcut for `<url>` in other commands.
`git fetch <remote> <branch>`	Fetches a specific `<branch>`, from the repo. Leave off `<branch>` to fetch all remote refs.
`git pull <remote>`	Fetch the specified remote's copy of current branch and immediately merge it into the local copy.
`git push <remote> <branch>`	Push the branch to `<remote>`, along with necessary commits and objects. Creates named branch in the remote repo if it doesn't exist.

ATLASSIAN

Visit atlassian.com/git for more information, training, and tutorials

GIT CONFIG

`git config --global user.name <name>`	Define the author name to be used for all commits by the current user.
`git config --global user.email <email>`	Define the author email to be used for all commits by the current user.
`git config --global alias. <alias-name> <git-command>`	Create shortcut for a Git command. E.g. `alias.glog "log --graph --oneline"` will set `"git glog"` equivalent to `"git log --graph --oneline.`
`git config --system core.editor <editor>`	Set text editor used by commands for all users on the machine. `<editor>` arg should be the command that launches the desired editor (e.g., vi).
`git config --global --edit`	Open the global configuration file in a text editor for manual editing.

GIT LOG

`git log -<limit>`	Limit number of commits by `<limit>`. E.g. `"git log -5"` will limit to 5 commits.
`git log --oneline`	Condense each commit to a single line.
`git log -p`	Display the full diff of each commit.
`git log --stat`	Include which files were altered and the relative number of lines that were added or deleted from each of them.
`git log --author= "<pattern>"`	Search for commits by a particular author.
`git log --grep="<pattern>"`	Search for commits with a commit message that matches `<pattern>`.
`git log <since>..<until>`	Show commits that occur between `<since>` and `<until>`. Args can be a commit ID, branch name, HEAD, or any other kind of revision reference.
`git log -- <file>`	Only display commits that have the specified file.
`git log --graph --decorate`	`--graph` flag draws a text based graph of commits on left side of commit msgs. `--decorate` adds names of branches or tags of commits shown.

GIT DIFF

`git diff HEAD`	Show difference between working directory and last commit.
`git diff --cached`	Show difference between staged changes and last commit

GIT RESET

`git reset`	Reset staging area to match most recent commit, but leave the working directory unchanged.
`git reset --hard`	Reset staging area and working directory to match most recent commit and **overwrites all changes** in the working directory.
`git reset <commit>`	Move the current branch tip backward to `<commit>`, reset the staging area to match, but leave the working directory alone.
`git reset --hard <commit>`	Same as previous, but resets both the staging area & working directory to match. **Deletes** uncommitted changes, and **all commits after** `<commit>`.

GIT REBASE

`git rebase -i <base>`	Interactively rebase current branch onto `<base>`. Launches editor to enter commands for how each commit will be transferred to the new base.

GIT PULL

`git pull --rebase <remote>`	Fetch the remote's copy of current branch and rebases it into the local copy. Uses git rebase instead of merge to integrate the branches.

GIT PUSH

`git push <remote> --force`	Forces the `git push` even if it results in a non-fast-forward merge. Do not use the `--force` flag unless you're absolutely sure you know what you're doing.
`git push <remote> --all`	Push all of your local branches to the specified remote.
`git push <remote> --tags`	Tags aren't automatically pushed when you push a branch or use the `--all` flag. The `--tags` flag sends all of your local tags to the remote repo.

Git vs GitHub – What's the Difference?

🛠️ *Git*

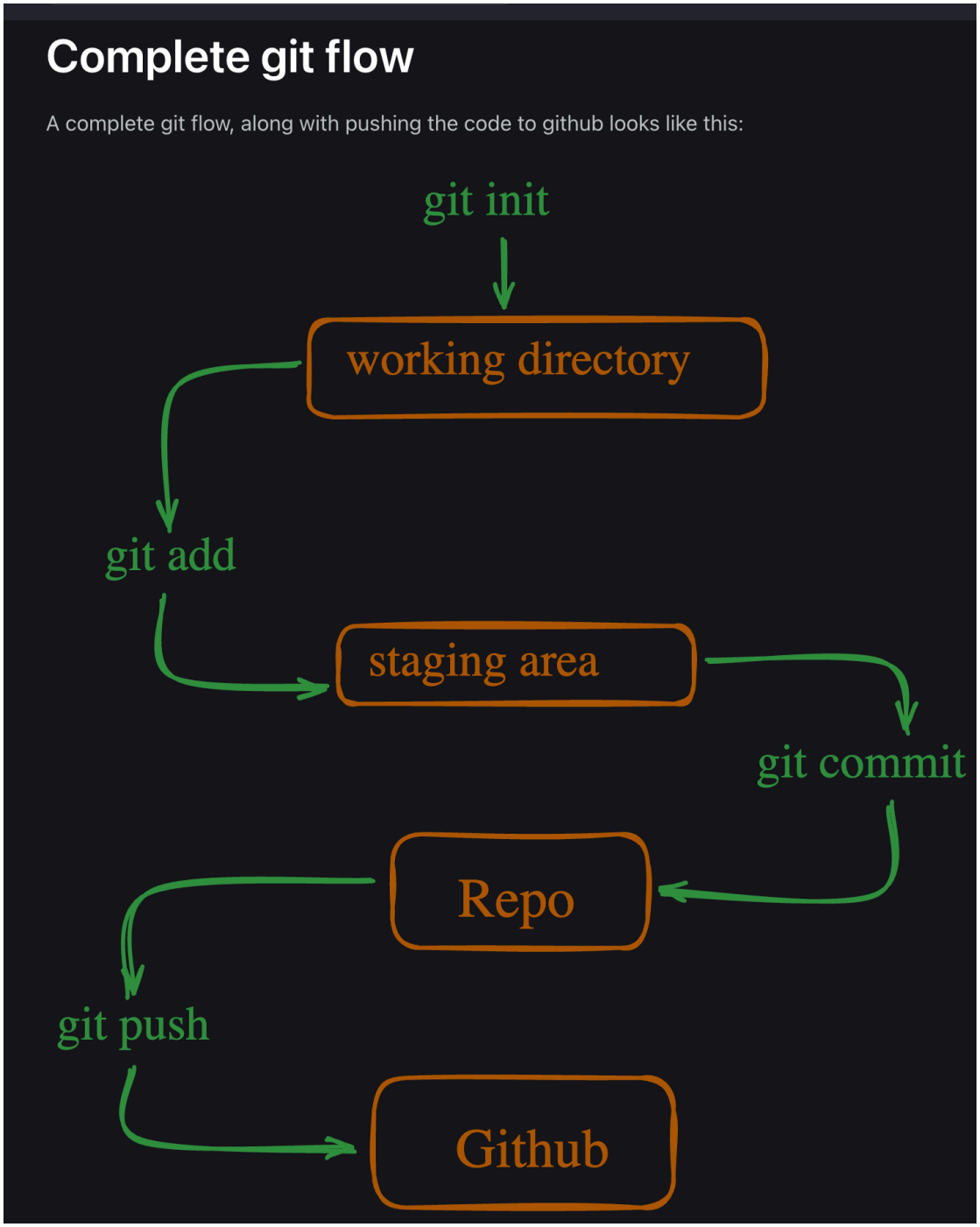
- Git is a **version control system**.
- It helps you **track changes** to your files and collaborate with others.
- Git is **free**, **open-source**, and can be installed on your local machine (Windows, macOS, Linux).
- Think of Git as the **tool** you use to save and manage versions of your code **on your own computer**.

🌐 *GitHub*

- GitHub is a **web-based hosting platform** for Git repositories.
- It lets you **store your Git projects online** and share them with others.
- It's a popular platform for **collaborating** on code and contributing to open-source projects.
- **Note**: GitHub is just one of many Git hosting services (others include GitLab, Bitbucket, etc.).

🎮 What is Version Control?

- A version control system (like Git) lets you **track the history of your code**, so you always know what changed, when, and by whom.
- It's like having **checkpoints in a video game** — you can move forward, go back, or recover from a mistake without losing progress.
- This is **crucial** in software development, especially when working in teams.



When you want to track a new folder, you first use init command to create a new repository. Then you can use add command to add the folder to the repository. After that you can use commit command to save the changes. Finally you can use push command to push the changes to github. Of course there is more to it but this is the basic flow.

#Git always recommends us whatever we write code, maybe 1000 lines of codes , 100s of features , don't add all of them at once. Add commits in such a way that one unit of work is done.

BRANCH –
Git switch <branchname>
Git switch -c <branchname> (not only switches but creates the branch also)
Git checkout <branchname> (branch should / must already exist to switch)

Stash -
Stash is a way to save your changes in a temporary location. It is useful when you want to make changes to a file but don't want to commit them yet. You can then come back to the file later and apply the changes. Conflicting changes will not allow you to switch branches without committing the changes. Another alternative is to use the git stash command to save your changes in a temporary location.

Git stash – we can do multiple stashed area also.
Git stash list – it lists all stashing areas.
Git stash apply – it applies whichever stack is at top of stack. (it saves in stash@{0}, stash@{1}, stash@{2} etc.)

Git stash pop – it applies the changes in our branch , and pops , we don't have to write git stash apply.

Git stash apply stash@{0} <branch-name> - we can apply stash to anyother specific branch also.

Git stash clear – it clears all the stashes

Revert --
Used to undo a committed change by creating a new commit that reverses the changes. bash Copy Edit git revert <commit>

Restore –
git restore . -----→Discards all unstaged changes (resets to last commit)
git restore --staged ----→Moves files from staged to unstaged (keeps changes

Reset –
git reset → Unstages files that were added via git add (like restore --staged)
git reset –hard <commit>→ Wipes everything: unstaged + staged changes
git reset --soft <commit> → Moves HEAD to the given commit, but keeps changes staged